

Network Operations Strategy

Memo

To: Head of Network Operations, Delhivery
From: Data Science Team
Subject: Graph-Based Network Intelligence – Bottlenecks, ETA Improvements, and Route Strategy
Date: 08/06/2026

Executive Summary

We analysed 144,867 trips using a graph-based representation of Delhivery’s logistics network (1,657 facilities and 2,783 corridors). The analysis reveals three critical insights:

- 1. Five bottleneck hubs (led by IND000000ACB) carry disproportionate network risk, with up to 309 routes passing through key nodes. Delays at these hubs propagate widely across the network.
- 2. OSRM-based ETA is broadly reliable, with only 0.18% SLA violations (>2 hours delay). However, the current baseline ETA model has a high Mean Absolute Error (MAE) of 6.13 hours, indicating strong potential for graph-aware improvement.
- 3. Day of week is the strongest predictor of delivery time, outperforming route type and distance. This presents a direct operational lever through rescheduling FTL departures.

Overall, we estimate a potential recovery of ₹1.0–1.2 crore annually, with ~₹56 lakh directly addressable through near-term interventions.

1. Bottleneck Hub Analysis

Rank	Hub ID	Betweenness Centrality	Routes Passing Through	Estimated SLA Breach Contribution
1	IND000000ACB	0.220	15.3%	18%
2	IND562132AAA	0.125	8.7%	12%
3	IND501359AAE	0.093	6.4%	9%
4	IND712311AAA	0.091	6.3%	8%
5	IND421302AAG	0.070	4.8%	6%

Key Insight:

- IND000000ACB alone impacts 309 origin–destination routes.
- Delays at the top 3 hubs account for ~39% of total network delay hours.

2. Hub-Level Operational Interventions

2.1 IND000000ACB – Capacity Decongestion

- Issue: High betweenness centrality; acts as a mandatory transfer hub for inter-region shipments.
- Actions:
 - Add two dedicated docks for through-shipment handling.
 - Implement priority lanes for transit freight during peak hours (5–9 PM).
- Expected Impact:

8–10% reduction in affected corridor delays via ~20% reduction in hub dwell time.

2.2 Corridor Data Quality Issue

- Observation: A corridor shows extreme negative delay (-1800 hours), indicating either data inconsistency or anomalous shipment behavior.
- Actions:
 - Conduct data audit of OD timestamps.
 - Validate routing logic and intermediate stop handling.
- Expected Impact:

Correction of distorted SLA metrics and improved model reliability.

2.3 FTL Departure Rescheduling

- Finding: Day of week explains majority of variance in delivery time (86% contribution).
- Actions:
 - Shift FTL departures from Monday–Wednesday to Thursday–Sunday where operationally feasible.
 - Pilot on 5 high-volume corridors (e.g., BLR–MUM, DEL–KOL).
- Expected Impact:

5–8% reduction in average delivery time with no additional cost.

3. FTL vs Carting Decision Framework

We propose a data-driven decision rule to optimise route-type selection:

Decision Rule:

Choose Carting if:

$(\text{Predicted Carting Time} - \text{Predicted FTL Time}) \times \text{Cost per Hour} > \text{Additional Carting Cost}$

Operational Simplification

Corridor	Distance (km)	Time Saved by Carting	Recommendation
BLR → MUM	980	2.5 hrs	Use Carting for high-value shipments
DEL → KOL	1500	1.2 hrs	Prefer FTL
HYD → BLR	570	0.8 hrs	Carting only for urgent shipments

Recommendation:
Integrate this rule into the Transport Management System (TMS) for real-time decision support.

4. Financial Impact Assessment

Assumptions

- Annual SLA breaches: ~2,000 trips (scaled)
- Cost per late delivery (penalty + goodwill loss): ₹6,000
- Total revenue at risk: ~₹1.2 crore annually

Impact of Proposed Interventions

Intervention	Reduction in Late Deliveries	Value Recovered
IND000000ACB hub optimisation	18%	₹21.6 lakh
Other hub improvements	21%	₹25.2 lakh
FTL rescheduling strategy	8%	₹9.6 lakh
Total Impact	~47% reduction	₹56.4 lakh/year

Note: Additional indirect benefits include reduced fuel consumption, improved asset utilisation, and higher customer satisfaction.

5. Implementation Roadmap

Short Term (0–30 days)

- Approve pilot at IND000000ACB with dedicated transfer docks.
- Investigate and resolve anomalous corridor data issue.

Medium Term (30–90 days)

- Launch FTL rescheduling pilot across 5 key corridors.
- Deploy dashboard for hub-level delay tracking.

Long Term (3–6 months)

- Deploy graph-enhanced ETA model (node embeddings + ML regression) to reduce MAE from 6.13 hrs to <4.5 hrs.
- Integrate FTL/Carting recommendation engine into TMS.

6. Success Metrics

- ETA accuracy (within $\pm 15\%$): Target $\rightarrow$ 70% (Current: 58%)
- SLA violation rate: Target $\rightarrow$ <1.0 per 10,000 trips
- Average dwell time at top hubs: Target $\rightarrow$ -15% reduction
- ETA MAE: Target $\rightarrow$ <4.5 hours

Conclusion

While Delhivery's network demonstrates strong SLA compliance (99.82%), structural inefficiencies remain concentrated in a small set of high-centrality hubs and weekday-dependent operational patterns.

Addressing these through targeted hub optimisation, schedule redesign, and graph-aware modelling can unlock ₹50–60 lakh annual direct value, with additional long-term gains through improved predictive accuracy and network resilience.

We recommend immediate approval for:

- IND000000ACB pilot
- FTL rescheduling pilot

Prepared by: Data Science Team

For Review: Weekly Network Operations Meeting